sealed.

PROJECT PROGRESS REPORT

Implementation & validation report

October 18, 2026 / Implementation snapshot 941a591

586	9	Live acceptance
backend tests passed	frontend tests passed	new calls & complete workflow pending

Prepared October 18, 2026. Implementation snapshot: `main` at `941a591`.

Sealed now connects dealer selection, guarded outreach, written-quote processing, negotiation, explicit buyer approval and written confirmation through a local dashboard and durable backend. Its complete synthetic workflow passes automated and browser checks. Individual integrations have also been exercised with live services. A fresh live acceptance of the checked call protocol and a complete live dashboard negotiation remain outstanding.

This report covers the work represented by the current repository and the recorded session evidence. Historical live tests are identified separately from synthetic tests; the report does not establish current vehicle availability or the current health of external services. Credentials, customer ceilings, raw messages and private provider conversation identifiers are excluded.

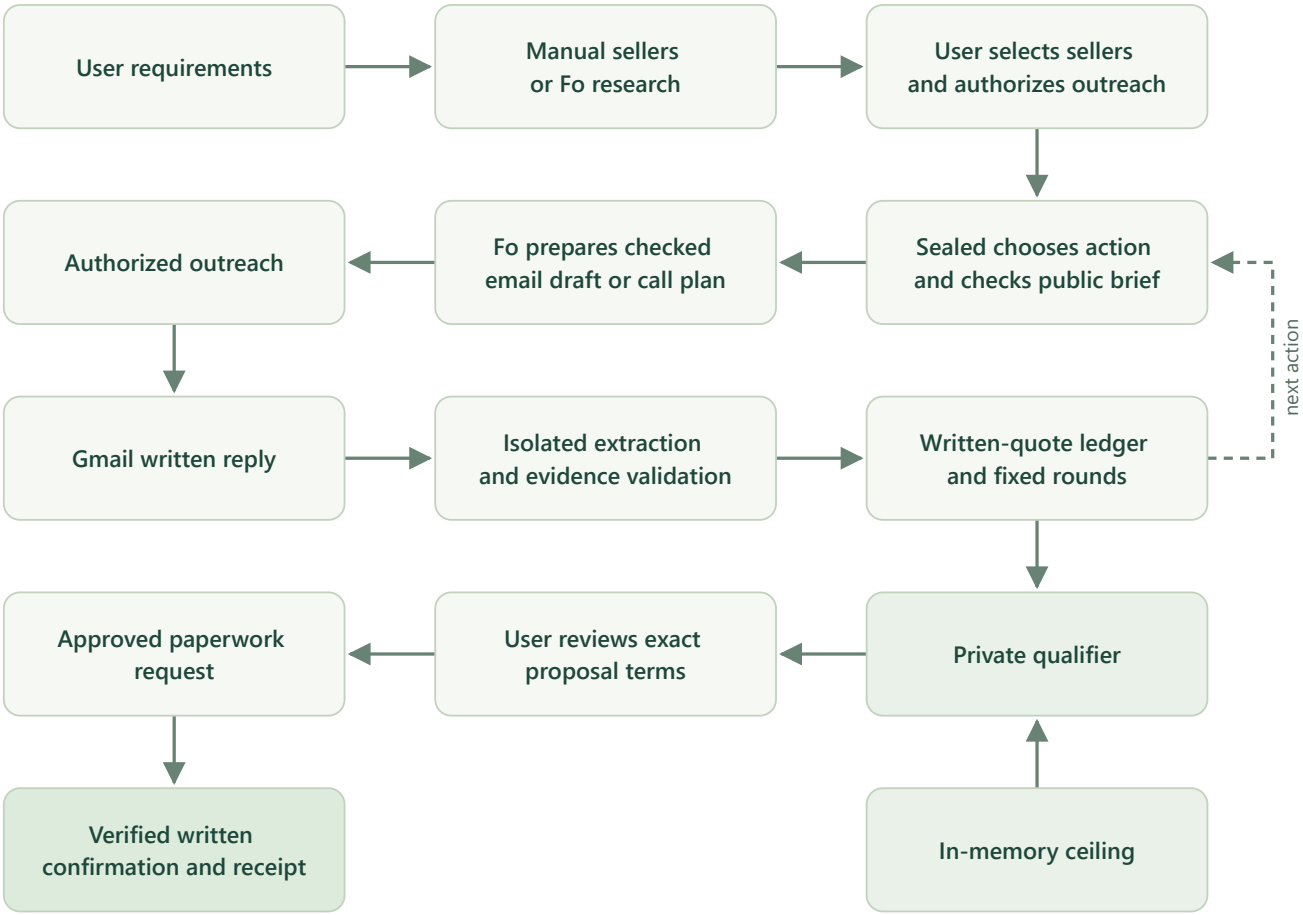
Product and architecture

The initial use case is purchasing one specified car from up to five sellers. The user supplies the specification, ZIP, deadline, public target and private ceiling. Sellers can be entered manually or selected from discovery results.

The implementation assigns distinct responsibilities:

Component	Responsibility	Main implementation
React dashboard	Intake, dealer selection, progress, evidence, approval and receipt	`frontend/src/`
FastAPI application	Local HTTP/SSE interface and worker lifecycle	`backend/app/`
Negotiation engine	Fixed rounds, quote authority, action selection and approval state	`backend/sealed/core/`
Private runtime	In-memory ceiling, secret checks and proposal qualification	`backend/sealed/private/`
Fo adapter	Checked email drafts, call plans, account serialization and provider state	`backend/sealed/fo/`

Component	Responsibility	Main implementation
Gmail ingestion and auditing	Authenticated written evidence, sent copies and reply linkage	`backend/sealed/ingest/`
Quote parser	Tool-free Claude extraction followed by deterministic evidence checks	`backend/sealed/parser/`
Coordinator	Polling, delivery, timers, guarded dispatch and recovery	`backend/sealed/service/`
Dealer discovery	Background research, public-source checks, distance filtering and saved selections	`backend/sealed/discovery/`



Public outreach and written evidence form the negotiation loop. The ceiling reaches only the private qualifier in this workflow diagram.

Fo chooses natural wording within an authorized brief. Sealed decides which action is permitted, which facts and prices it may use, and when to stop for approval or review. Claude extracts seller terms without outreach tools.

Negotiation logic and private ceiling

We implemented and connected the fixed two-round strategy:

1. Request a written out-the-door total for the specified car. A phone-only seller can receive an authorized call requesting a written quote.

2. Ask once for clarification when the written reply is insufficiently firm. After fixed waits, eligible silent sellers can receive a call nudge.
3. During round two, tell higher-priced sellers the lowest genuine competing written total. Ask a seller holding the lowest total for the public target when that target is lower than its quote.
4. At the end of the rounds, privately qualify the available written offers and present a proposal or request human review.
5. Require the user's exact-term approval before requesting purchase paperwork. Complete only after the seller's verified written confirmation.

For the controlled example, a $100 written quote and $90 public target produce a $90 counter. The system does not repeatedly increase its offer or bargain indefinitely until the target is reached. A qualifying final offer can be above the target and still require the user's approval.

The ceiling is held in backend memory and excluded from public job records, events, checkpoints and outreach. The strategy has no ceiling parameter. The private qualifier uses it to decide whether to propose an offer; the checker uses it for disclosure detection. The user's dashboard obtains it through a dedicated private endpoint and keeps it out of browser storage. Restarting the backend clears that memory, so further authorized job actions can require restoration. Observing an already authorized pending call can continue without restoring the ceiling.

Fo connection and email delivery

We implemented the MCP adapter with one owned async session behind synchronous methods, reconnect/cleanup behavior, OAuth refresh support and direct API-key authentication. A configured `FO_API_KEY` takes precedence over OAuth. Key rejection does not silently switch accounts through OAuth fallback. Credentials remain in ignored local configuration and are excluded from frontend settings.

New emails use a structured public brief rather than requiring Fo to deliver an exact template. Fo must return a draft before sending. Code checks the recipient, CC, subject, AI disclosure, specification, authorized price roles, commitments and action scope. Ordinary compliant signatures and rewording are allowed.

We added Gmail auditing of the actual sent copy. A reviewed draft alone is not delivery evidence. The auditor checks routing, authentication and content against the brief and saves the real reply/thread bindings. An uncertain or mismatched result retains its review state instead of triggering another send.

Recorded live evidence includes an initial controlled email and a later $90 counter in the original test conversation. The counter passed draft review and all 26 actual-copy audit rules. The earlier exact-template test retained its signature-only mismatch as history; reviewing it did not erase that evidence or resend the email. API-key authentication was separately checked with two read-only conversation-list requests through the same client.

See **Fo adapter**, **email briefs** and **sent-copy auditing**.

Written replies, parsing and durable processing

Gmail IMAP supplies the written source evidence. Ingestion associates replies using trusted subject/reply/thread bindings, checks the configured seller and Gmail authentication evidence, strips common quoted history, and stages messages with durable deduplication. An unknown sender or subject tag alone does not establish a phone-only dealer's identity.

The isolated Claude parser receives cleaned reply text and limited public parsing context. It receives neither the target nor ceiling and has no acting tools. Its structured output includes evidence spans; code checks amounts, all-in terms, specification, dates, validity and itemized costs against the actual text. Ambiguous or unsupported evidence cannot become firm merely because the model labels it firm.

A user-authorized live parser test extracted the $95 sample's total, all-in terms, availability and validity. Its missing specification confirmation correctly prevented a firm result. Durable handoff then applied the cached real reply once and queued one clarification. Immediate/restart replay did not create duplicate quotes or outreach; that handoff test sent nothing.

We connected ingestion and parsing to SQLite checkpoints, events and a guarded outbox. Reply effects commit before inbox acknowledgment. Outbound claims commit before external dispatch. Historical or superseded replies cannot silently replace newer authoritative terms, and changed seller terms invalidate stale approval. An ambiguous external outcome is preserved for reconciliation.

See **Gmail ingestion**, **quote parser** and **durable delivery**.

Dealer discovery

We added **Find dealers** to the intake form. In live mode, Fo researches public dealership websites using the item, specification, ZIP and 20-mile radius. Target, ceiling and buyer identity are excluded from the discovery request schema. Research permission is separate from seller-outreach permission.

The backend independently fetches cited public pages, checks literal business and contact evidence, removes duplicates and filters distances using ZIP centroids. It restricts website fetches to public HTTPS destinations, bounds response sizes and timeouts, and does not execute page scripts. Distances are approximate straight-line ZIP-centroid distances, not driving distances. Public postal lookup uses Zippopotam.us with the ZIP only.

Research runs in a dedicated worker with saved progress and one-hour completed result caching. Selected contact details are checked against saved provenance, requirements and freshness before job creation. Phone-only candidates remain possible; a website listing never proves current stock.

Recorded Fo research returned real candidates, and an integrated search retained a source-checked phone-only dealer while excluding candidates that failed checks. Those historical results do not establish current inventory. Synthetic discovery returns three explicitly labeled fictional candidates without Fo or web requests. We also clarified blocked-search messages and allowed explicit resubmission of a search that never reached Fo once its blocking account reservation is reviewed. Attempted uncertain searches are not blindly repeated.

Fo's research-only restriction is an instruction boundary. The MCP connection does not provide a mechanically isolated research-only tool environment.

See **dealer discovery** and **historical probe**.

Calls, identity and result tracking

An earlier real dealer-call attempt exposed several problems: an acknowledgment was treated as unclear, the job lacked a useful progress state, and Fo's reported introduction used an account-owner identity instead of the requested buyer. The call reached voicemail and produced no quote. Its possible follow-up activity was reviewed administratively, the specific reservation was released with retained evidence, and the cancelled negotiation stayed cancelled.

We implemented a new call lifecycle in response:

- Before dialing, require a strict plan matching the buyer, phone, exact opening, public brief, request identity and one-call limit. The plan must state that no call has started and no follow-up has been scheduled.
- Issue one separate dialing permission only after the plan passes checks.
- Save the pending attempt durably. Read its existing conversation to observe progress; do not repeat dialing permission after a timeout or restart.
- Require a scoped final report with outcome, introduction, follow-up declaration and complete reported transcript. Check identity, unsupported numbers, commitments, payment topics and follow-up promises.
- Publish `call.pending` and `call.completed`, with outcomes such as connected, voicemail, no answer or failed. Missing usable results after the 30-minute bound require review.

The dashboard now shows the pending state, dealer status and final reported outcome. A no-answer result without an introduction is labeled as having no spoken introduction. Spoken prices do not enter the firm written-price ledger. Raw call transcripts remain in ignored local provider state rather than public call events.

Observation can survive restart, absent private state and job cancellation. Cancellation stops queued Sealed actions; it cannot recall an external call. Other account outreach waits while a call remains pending. A narrowly scoped operator-review CLI can release a reviewed uncertain reservation without resending, relabeling history or resuming the job.

The new protocol passes controlled automated tests, but has not yet completed a fresh live call acceptance. Its final evidence is Fo-reported, without independent voice or scheduler attestation. Code checks do not mechanically control every action Fo could take internally.

See **call lifecycle** and **operator-reviewed recovery**.

Coordinator, API and dashboard

The background coordinator now connects polling, authorized parsing, reply delivery, timers, dispatch, approval and written confirmation. The API owns a single account executor and worker lifecycle. Discovery uses a separate worker but shares the Fo account reservation. Process locking prevents a competing live worker from operating against the same state.

The dashboard supports job creation/listing, seller selection, starting a job, separate research/call/model-processing permissions, SSE activity, private-state restoration, exact-term approval, rejection, cancellation and final receipts.

We incorporated Renu's dashboard work: per-dealer histories, review alerts, approval evidence, itemized fee comparison, a pinned summary, next-step card, tabs and ceiling controls. Subsequent updates added call lifecycle activity, clearer discovery errors and a small round review dot in place of the stretched orange indicator caused by a CSS class collision.

Local startup problems were also addressed/documented: use the project virtual environment from `backend/`, run one API instance, and restart it after backend or environment changes. The `coordinator_already_running` error indicates an existing worker lock; starting duplicate backend processes is not a recovery step.

The API remains a local application without user authentication. It enforces loopback access, local Host checks, an exact dashboard origin, JSON POSTs and no-store responses. These controls do not constitute a hosted multi-user design.

See **dashboard operations**, **API contract** and **coordinator**.

Verification evidence

Check	Recorded result	What it establishes
Full backend suite before `941a591`	586 passed	Engine, integrations with test doubles, privacy, recovery, API and call lifecycle behavior
Frontend suite	9 passed	Event-to-dashboard transformations and activity behavior
Frontend production build and lint	Passed	TypeScript compilation, bundling and configured lint checks
Latest targeted backend run	67 passed	Call lifecycle, coordinator and job API checks
Synthetic coordinator acceptance	Completed at $90 from a $100 baseline	Explicit simulated approval, written confirmation and restart-safe replay; zero live requests
Mixed email/call acceptance test	Passed	Call nudge followed by written quote, counter, approval and written confirmation
Earlier full browser simulation	Completed	Intake, event updates, approval and receipt through the rendered dashboard
Latest Chrome progress/result check	Passed	Pending call card, dealer history, final no-answer event and final negotiation receipt
Mobile layout check	Passed for tested views	No horizontal overflow at the tested 390-pixel width
Credential scan	Passed across 171 repository files	Actual configured secrets and detected key patterns absent from included files
Git whitespace check	Passed	No reported whitespace errors in the committed changes

The latest browser check used synthetic fetch/SSE responses against the production frontend build. It verified automatic updates without refreshing, no spoken-price quote, final total/savings/evidence, zero browser exceptions and no unexpected API requests. Its $90 receipt is a fixture, not a completed real purchase. Private screenshots and its report remain under ignored `.sealed/`.

Historical live evidence covers individual email, Gmail, parser, Fo authentication and research operations. It does not establish one uninterrupted live dashboard run from discovery through purchase confirmation.

Repository delivery

Commit	Scope
`803538f`	Durable negotiation backend, dashboard and Fo discovery checkpoint
`e2923b4`	Integration with the existing remote history
`cdc6bb6`	Merge of Renu's dashboard changes
`941a591`	Checked call lifecycle, API-key authentication, discovery recovery, UI fixes, tests and docs

The latest commit contains 38 changed files, with 1,504 insertions and 27 deletions. At report preparation, local `main` is one commit ahead of the locally recorded `origin/main`; `941a591` has not been pushed. No fresh remote fetch was performed for this report. The source working tree was clean before creating this report; this report is a new, uncommitted documentation file.

Local credentials, OAuth tokens, runtime databases, cached messages, provider state and screenshots remain ignored. The tracked `.env.example` contains blank credential fields.

Remaining work and readiness

Priority	Next task	Acceptance evidence needed
1	Run a consenting live call through the new protocol	Correct buyer and phone, one dialing permission, visible pending status, usable final report, no reported follow-ups
2	Run a complete controlled live dashboard negotiation	Real authenticated quote, guarded counter, explicit approval, exact written confirmation and receipt
3	Verify recovery during that controlled live run	Existing attempt observed after restart; no duplicate outreach; private ceiling restored before new actions
4	Finish/document phone-only seller identity onboarding	A written reply gains trusted seller attribution; an unknown email alone cannot establish it
5	Exercise operator handling of uncertain results	Reviewed release keeps evidence, preserves cancellation and does not imply permission for another action
6	Reconcile older documentation milestones	Remove obsolete statements that parsing, delivery or coordinator wiring are unfinished, and align older architecture call descriptions with the current call contract
Later, if hosted deployment is required	Design and validate hosted operation	Authentication, per-user/account isolation, deployment configuration, secrets management and operational recovery suited to that deployment

Readiness assessment: the local implementation is ready for a controlled demonstration and the next supervised live acceptance. The evidence does not yet support calling the full live workflow production-ready. The highest-value next step is the consenting one-call acceptance, followed by the complete controlled written-quote-to-receipt run.